

Geophysical Research Letters

RESEARCH LETTER

10.1029/2020GL087481

Key Points:

- Titan's ice shell is likely capped by a methane clathrate crust that has thermal conductivity 1 order of magnitude smaller than water ice
- The clathrate crust significantly reduces the stagnant lid thickness thus increasing the potential for the ocean-surface exchange
- The insulating effect of the clathrate crust makes convection much less efficient thus limiting the freezing rate of the deep ocean

Supporting Information:

- Supporting Information S1

Correspondence to:

K. Kalousová,
klara.kalousova@mff.cuni.cz

Citation:

Kalousová, K., & Sotin, C. (2020). The insulating effect of methane clathrate crust on Titan's thermal evolution. *Geophysical Research Letters*, 47, e2020GL087481. <https://doi.org/10.1029/2020GL087481>

Received 11 FEB 2020

Accepted 2 JUN 2020

Accepted article online 9 JUN 2020

The Insulating Effect of Methane Clathrate Crust on Titan's Thermal Evolution

Klára Kalousová¹  and Christophe Sotin² 

¹Department of Geophysics, Faculty of Mathematics and Physics, Charles University, Prague, Czech Republic, ²Jet Propulsion Laboratory, California Institute of Technology, Pasadena, CA, USA

Abstract Saturn's largest moon Titan is unique for its dense, methane-rich atmosphere, with ongoing complex organics chemistry, and the hydrocarbon lakes and seas on its frigid surface. Titan also harbors a subsurface water ocean, and its ice shell is likely capped by a thick crust of low-conductivity methane clathrates that are very stable under Titan's surface conditions. Here, we investigate the effect of this insulating clathrate crust on convection in the ice shell. Without the clathrate crust, convection occurs below a stagnant lid that is a few tens of kilometers thick and limits the exchange between the surface and the deep ocean. We show that a 5–10 km thick clathrate crust reduces the stagnant lid thickness about threefold thus increasing the potential for the ocean-surface exchange. Moreover, the insulating effect of clathrates dramatically decreases the amount of extracted heat thus limiting the freezing rate of Titan's ocean.

Plain Language Summary The Cassini mission has revealed that Titan is an active icy moon with organic molecules forming in its atmosphere and a climate characterized by methane rains. Titan also harbors a deep ocean that decouples the ice shell from the inner solid layers. Methane and water ice react to form clathrate hydrate that acts as a thermal insulator. Previous one-dimensional models of Titan's thermal evolution suggested the formation of a clathrate crust early in Titan's history, before the ice shell started to crystallize. Here, we follow with two-dimensional numerical simulations of thermal convection in the ice shell that take into account the insulating effect of a clathrate crust. We show that the lithosphere thickness is reduced significantly thus providing a pathway for the exchange between the surface and the ocean. In addition, the amount of heat transferred by convection is reduced, which limits the freezing rate of Titan's ocean. The clathrate crust destabilization by impacts or cryovolcanism can also provide the methane needed to replenish the atmosphere. Titan's methane cycle significantly influences the ice shell structure and its dynamics thus playing a role similar to water on the Earth which not only shapes its surface but also affects its interior dynamics.

1. Introduction

Titan is the only moon in the solar system with a dense atmosphere, the only object we know, besides the Earth, that has liquids at its surface (Stofan et al., 2007), and a world of organic molecules produced in the atmosphere by photolysis of methane and further reactions with nitrogen (Hayes et al., 2018; Vuitton et al., 2007). The Cassini mission demonstrated that Titan has a deep ocean below an ice shell between 50 and 100 km thick (Baland et al., 2014; Béghin et al., 2012; Iess et al., 2012). After Cassini's plunge into Saturn's atmosphere, several key questions remain: How can the deep liquid ocean be maintained? Can the surface organic material be transferred into the ocean? How is Titan's atmosphere replenished in methane which has a lifetime of only 20–30 Myr (Wilson & Atreya, 2000; Yung & Allen, 1984)? All these questions point to the ability of the ice shell to (i) extract heat from Titan's interior, which leads to freezing of its ocean, and (ii) transport chemical compounds thus providing means for the exchange between the surface and the interior.

To explain the observed methane abundance in Titan's atmosphere, Tobie et al. (2006) proposed its replenishment via episodic outgassing of methane stored in the form of clathrate. They predicted that a methane clathrate crust formed on top of Titan's ocean before water ice started to crystallize. Methane clathrates could have accreted in Titan's core (Lunine & Stevenson, 1987) and remained stable throughout differentiation (Loveday et al., 2001) or formed in the core either during the core overturn (Tobie et al., 2006) or later due to the cracking of organic matter (Miller et al., 2019; Néri et al., 2020). For a power between 0.5 and

1 TW coming from Titan's core before ice crystallization (Tobie et al., 2006), the clathrate crust had to be between 15 and 7.5 km thick, respectively, to prevent significant ice crystallization (see supporting information [SI] for more details). According to the model of Tobie et al. (2006), subsolidus convection started ~ 0.5 –1 Gyr ago and led to fast water ice crystallization with the result being an ice shell capped by a clathrate crust.

Another indication on the presence of methane clathrates at Titan's surface is the methane cycle. Models of Titan's present-day climate predict methane rains mostly at the poles during summer (Mitchell & Lora, 2016; Schneider et al., 2012) with precipitation rates of 1–2 mm/day. The lakes discovered by the Cassini mission at the north pole are also dominated by methane (Mastrogiuseppe et al., 2016). The liquids present on Titan's surface can percolate into the subsurface to form an “alkanofer”—a porous ice layer filled with hydrocarbons (Hayes et al., 2008; Mousis et al., 2014; Turtle et al., 2018). As recently shown by Vu et al. (2020), clathrates will form by reaction of liquid hydrocarbons with water ice. The reaction will be complete on seasonal time-scales even at Titan's surface temperature thus suggesting that only a negligible amount of pure water ice is mixed with the initial clathrate crust. Note also that clathrates are extremely stable under Titan's present-day surface conditions.

Since the thermal conductivity of clathrates is about an order of magnitude lower than that of water ice at Titan's surface temperature (Sloan et al., 2007), a clathrate crust would reduce the amount of heat extracted from Titan's interior. This scenario was already investigated by Tobie et al. (2006) who used scaling laws to describe heat transfer through Titan's ice shell. In this study, we perform numerical simulations of thermal convection in an ice shell capped by an insulating methane clathrate crust to describe its effect on Titan's thermal evolution.

2. Model

We solve thermal convection of viscous fluid (ice) in a two-dimensional Cartesian box assuming incompressible Boussinesq approximation of the conservation laws. We include the strongly nonlinear viscosity of ice as well as highly variable thermal conductivity. The governing equations can be written as

$$\nabla \cdot \mathbf{v} = 0, \quad (1a)$$

$$0 = -\nabla p + \nabla \cdot (2\eta\dot{\epsilon}) + \rho g \alpha (T - T_0) \mathbf{e}_z, \quad (1b)$$

$$\rho c_p \left(\frac{\partial T}{\partial t} + \mathbf{v} \cdot \nabla T \right) = \nabla \cdot (k \nabla T) + Q, \quad (1c)$$

with $\mathbf{v}$ the ice velocity, p the dynamic pressure, η the ice viscosity, $\dot{\epsilon}$ the strain rate tensor, ρ the ice density, g Titan's gravity, α the ice thermal expansivity, T and T_0 the actual and reference ice temperature, respectively, $\mathbf{e}_z$ the unit vector in vertical direction, c_p the ice isobaric heat capacity, t the time, k the thermal conductivity, and Q the tidal heating rate.

For viscosity, we use the stress-dependent composite law of Goldsby and Kohlstedt (2001). The main parameter controlling the actual viscosity value is the grain size d —we consider values between 0.5 mm (leading to smallest viscosity) and 2 mm. Thermal conductivity of water ice is indirectly proportional to temperature (Hobbs, 1974) and varies between $2.32 \text{ W (m K)}^{-1}$ at the interface with the ocean and $5.66 \text{ W (m K)}^{-1}$ at Titan's surface. To include the insulating effect of clathrates, the thermal conductivity is reduced to $k = 0.5 \text{ W (m K)}^{-1}$ (Sloan et al., 2007) in the crust of thickness h_c . Finally, for the tidal dissipation we use the approach described in Tobie et al. (2003). The whole system is implemented in the open source Finite Element Method library FEniCS (Alnæs et al., 2015; Logg et al., 2012). More details and the values of the particular parameters can be found in the SI.

3. Results

We performed a series of numerical simulations of thermal convection in Titan's shell assuming it is 100 km thick. All simulations reached a statistical steady state. We first describe the convection pattern and heat transfer efficiency through Titan's shell as a function of the clathrate crust thickness h_c . Then, we investigate the effect of ice grain size d (viscosity) and tidal dissipation in the shell. Finally, we present implications for Titan's thermal evolution.

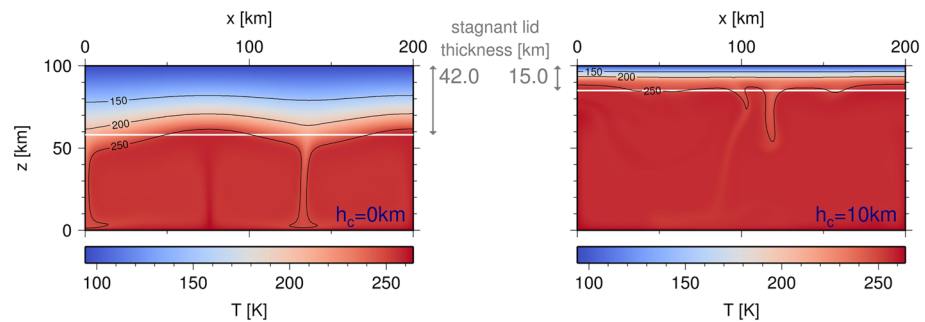


Figure 1. Presence of a clathrate crust brings warm material close to the surface. Comparison of temperature fields assuming a clathrate crust 0 km (left) and 10 km (right) thick. Grain size is 1 mm in both cases. The white line indicates the stagnant lid, and the black contours correspond to $T = 150, 200$, and 250 K.

3.1. Effect of a Clathrate Crust On the Ice Shell Dynamics

We start with a detailed comparison of the results of two simulations—one without clathrates and one with a 10 km thick clathrate crust (Figure 1). In both cases, the grain size is 1 mm. If clathrates are not present, we observe the formation of a steady flow pattern with a stagnant lid more than 40 km thick (Figure 1, left). The base of the stagnant lid roughly corresponds to the average depth of the 250 K isotherm. The inclusion of a 10 km thick crust of low-conductivity clathrates reduces the thickness of the stagnant lid almost threefold to 15 km (Figure 1, right). Due to a thinner stagnant lid, the warm convecting material is brought much closer to Titan's surface. The resulting convection pattern is also rather chaotic due to a significantly thicker layer of warm material where the viscosity is $\sim 1.5 \times 10^{15}$ Pa s with diffusion creep controlling the ice deformation. Viscosity is thus only temperature dependent in the convective domain.

Figure 2 shows a comparison of the horizontally averaged temperature profiles for simulations with different clathrate crust thickness ($h_c = 0, 5, 10$, and 15 km, left to right) and grain size of 1 mm. The clathrate crust acts as an insulator owing to its very small thermal conductivity. As described above, its presence dramatically reduces the thickness of the stagnant lid (h_{sl} , light blue boxes) from 42 km (no clathrates) to 15 km (10 km of clathrates). Interestingly, assuming even thicker clathrate crust ($h_c = 15$ km) leads again to a slightly thicker stagnant lid ($h_{sl} = 18.5$ km) because the surface heat flux q_s decreases (dashed cyan lines).

Apart from the stagnant lid thinning, the presence of clathrates also increases the temperature of the convective interior T_c . For the 1 mm grain size, the convective temperature T_c increases from 254 K ($h_c < 5$ km) to 258 K ($h_c = 15$ km). In other words, the distance from the melting temperature of pure ice decreases by 4 K. Combined with the thinner stagnant lid, the inclusion of a clathrate crust results in bringing a hot material close to Titan's surface.

Another effect of increasing the convective temperature is that the temperature contrast $\Delta T_\delta = T_b - T_c$ (with T_b the ocean temperature) in the hot thermal boundary layer δ (TBL, light red boxes in Figure 2) decreases. As a result, the buoyancy of warm ice at the interface with the ocean is reduced and convection becomes less efficient. Therefore, a significantly smaller amount of heat is extracted from the ocean—the basal heat flux q_b (red dashed lines) decreases by almost 60% from 13.1 mW m^{-2} for the ice shell without clathrates to 5.4 mW m^{-2} with $h_c = 15$ km.

3.2. The Effect of Viscosity and Tidal Dissipation

A few parameters that are not well constrained may modulate the effect of the clathrate crust on the ice shell dynamics. Varying the grain size d (and therefore the viscosity) does not qualitatively change the above findings. The presence of the clathrate crust increases the convective temperature T_c by up to 5 K for h_c ranging from 0 to 20 km and all investigated grain sizes (Figure 3a). The corresponding decrease of ΔT_δ by up to 5 K represents a significant reduction of the buoyancy of hot material at the ocean interface (Figure 3b). The minimum stagnant lid thickness (Figure 3c) changes from 11.5 km for 0.5 mm grain (smallest viscosity) to 19 km for 2 mm grain, but as the thickness of the clathrate crust h_c increases, the difference in the stagnant lid thickness h_{sl} for different grain sizes d becomes less pronounced. Similarly, the difference in the basal heat flux q_b for different grain sizes becomes less pronounced with the increasing clathrate crust

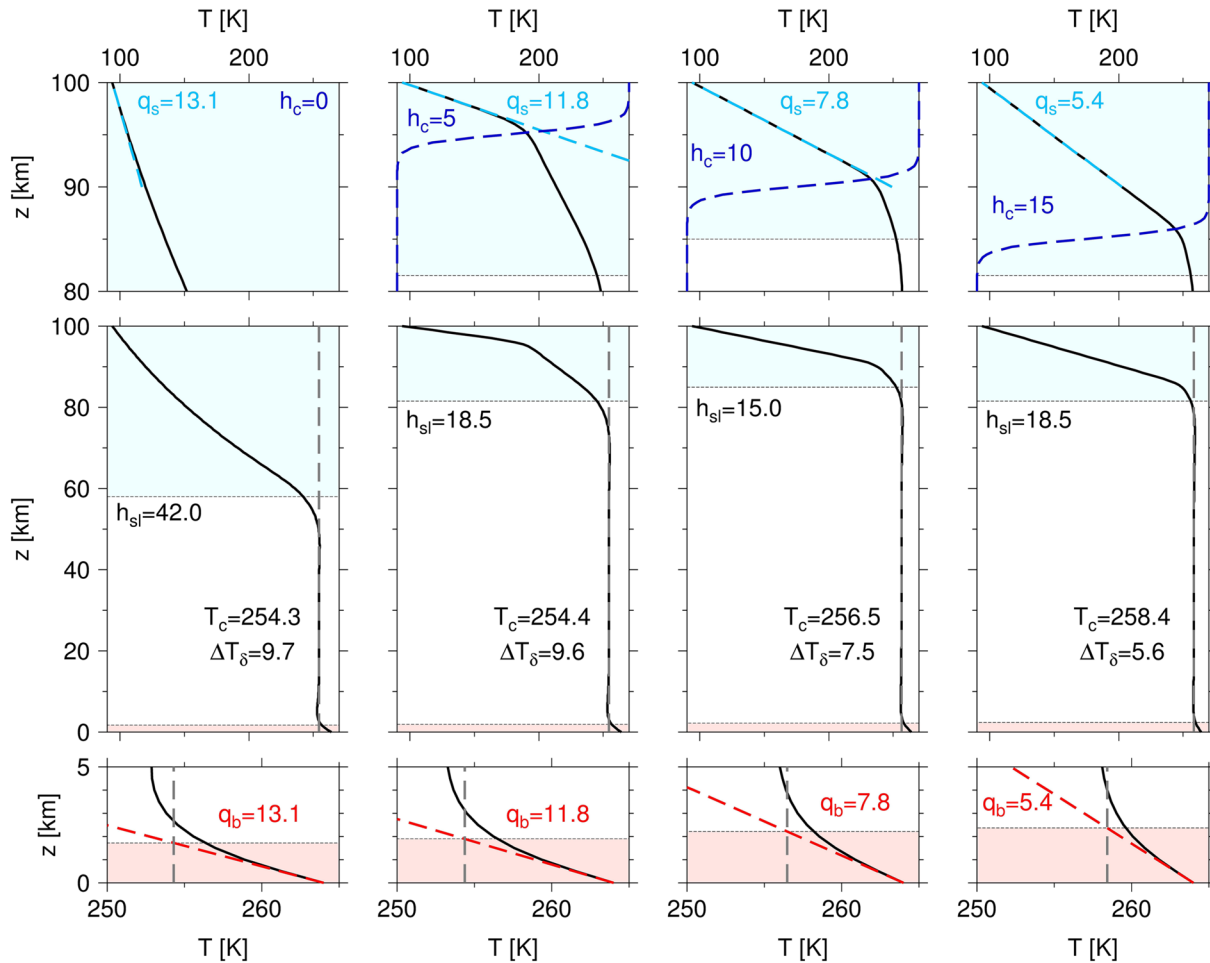


Figure 2. Clathrate crust decreases stagnant lid thickness and basal heat flux. Horizontally averaged temperature profiles (black) for clathrate crust thickness $h_c = 0, 5, 10,$ and 15 km (left to right). Top 20 km (top panels), the whole 100 km thick shell (middle panels), and bottom 5 km (bottom panels). The dashed lines indicate the clathrate crust thickness (h_c [km], dark blue), the surface heat flux (q_s [mW m^{-2}], cyan), the convective temperature (T_c [K], gray), and the basal heat flux (q_b [mW m^{-2}], red). The colored boxes show the stagnant lid (h_{sl} [km], light blue) and the hot thermal boundary layer (δ , light red).

thickness h_c (Figure 3d). For all grain sizes, the ratio of convective to conductive heat flux q_d (Figure 3d, black dashed line) varies between 1.5 and 3.5.

Tidal dissipation in the ice shell can also reduce the amount of heat that can be extracted from Titan's interior. Its amplitude depends on eccentricity, obliquity, and the imaginary part of the tidal Love number which strongly depends on the ice viscosity. A significant tidal dissipation would have dampened Titan's large eccentricity, but the dampening would only have started after the ice shell has thickened enough and started convecting as demonstrated by Tobie, Grasset, et al. (2005) and Tobie et al. (2006). Following Tobie, Mocquet, et al. (2005), we estimate the maximum volumetric tidal heating rate Q_m in Titan's shell to be $\sim 5 \times 10^{-8} \text{ W m}^{-3}$ (see SI for more details). Note that this is much less than tidal dissipation rates in Europa's ice shell (Tobie, Mocquet, et al., 2005) or in Enceladus' porous core (Choblet et al., 2017) which explains the presence of an ocean in these two bodies. Nevertheless, this effect may still not be negligible. Results for three clathrate crust thicknesses, $h_c = 0, 5,$ and 10 km, and 1 mm grain size are reported in Figure 3 (orange squares). In the absence of clathrates ($h_c = 0$ km), the inclusion of tidal heating results in the convective temperature T_c increase and the ΔT_δ decrease by about 1.5 K. The reduction of the basal heat flux extracted from the ocean is almost 20%. When a 5 and 10 km thick clathrate crust is included in the simulation together with the tidal heating, the increase in convective temperature (decrease in ΔT_δ) with respect to simulation with tidal heating only (without clathrates) is about 1.5 and 5 K, respectively.

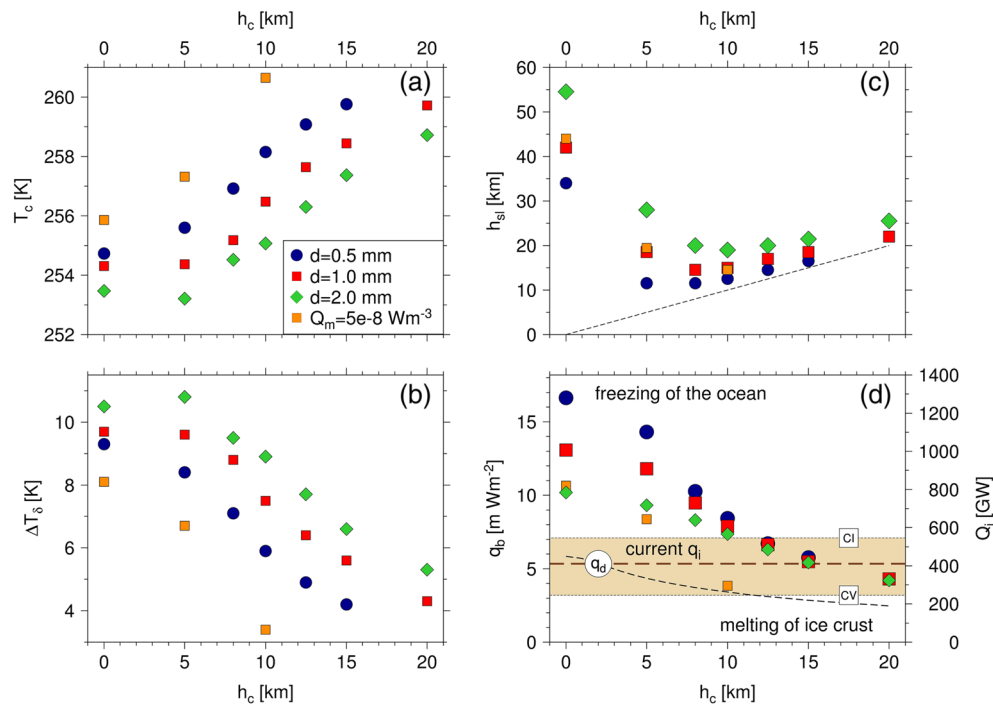


Figure 3. Clathrate crust prevents the freezing of Titan's ocean. (a) Convective temperature T_c , (b) TBL temperature contrast ΔT_δ , (c) stagnant lid thickness h_{sl} , and (d) basal heat flux q_b extracted from the ocean. All quantities are plotted as a function of the clathrate crust thickness h_c . Basal heat flux can be compared to the estimated range of current heat flux from Titan's interior q_i (light brown box, see SI for more details) and internal heat given by Chen et al. (2014) (brown dashed line). The conductive heat flux q_d through the shell (black dashed line) is plotted for comparison with the convective heat fluxes determined in this study.

Consequently, the basal heat flux is reduced by $> 20\%$ for $h_c = 5$ km and $> 60\%$ for $h_c = 10$ km. Tidal dissipation in the ice shell combined with the presence of the clathrate crust thus significantly limits the basal heat flux q_b that can be extracted from Titan's interior.

3.3. Implications for Titan's Ocean

As demonstrated above, an insulating clathrate crust reduces the amount of internal heat that can be extracted through Titan's ice shell. If the heat flux q_b extracted from the interior is larger than the internally generated heat q_i , Titan's interior cools and its ocean freezes. Internal heat sources include the decay of long-lived radioactive elements present in the silicate core and tidal dissipation due to Titan's eccentricity and obliquity in the core and the ocean (Chen et al., 2014). Using the concentrations of radioactive elements measured on a variety of carbonaceous chondrites (Braukmüller et al., 2018) and the energy per unit mass released during the decay of each element (Van Schmus, 2013), we estimate the amount of radioactive power to range between 250 and 545 GW (see SI for more details) which is in good agreement with the value of 411 GW reported by Chen et al. (2014) (dashed brown line in Figure 3d) and is orders of magnitude larger than tidal dissipation in the core (Tobie, Mocquet, et al., 2005). Similarly, tidal dissipation in the ocean is not expected to contribute significantly into the heat budget (Chen et al., 2014). The corresponding heat flux q_i coming to Titan's ice shell from the interior is thus between 3.2 and 7.1 mW m⁻² (light brown box in Figure 3d).

Our results show that if the ice shell is convecting, the long-term existence of a liquid ocean is best explained by a thick clathrate crust (> 10 km) or a combined effect of tidal dissipation and a moderately thick clathrate crust (~ 7 – 10 km). In these cases, the basal heat flux q_b is comparable with the internal heating q_i (Figure 3d). On the other hand, the inclusion of tidal dissipation alone results in extraction of more heat than is supplied from the interior (Figure 3d) and is thus not sufficient to prevent Titan's ocean from freezing. Alternatively, the ocean may be explained if the ice shell is cold and conductive (see the dashed line in Figure 3d) which would require larger ice grains than considered in this study or significant amount of ammonia in the ocean (see section 4 for more details).

4. Discussion and Conclusions

The effect of an insulating clathrate crust on heat transfer through a conductive shell is well understood—for a given shell thickness, the amount of internal heat that can be extracted is reduced when clathrates are present (see the dashed line in Figure 3d). If heat is transferred by convection, the effect of clathrates cannot be predicted by scaling laws developed in the stagnant lid convection regime without clathrates. Results of convection simulations with temperature-dependent viscosity but without clathrates (Deschamps & Sotin, 2001) have shown that the hot instabilities which form in the TBL determine the overall convection pattern. The TBL heat flux—given as $k\Delta T_{\delta}/\delta$ —thus constrains the heat flux through the ice shell. Since ΔT_{δ} is proportional to the viscous temperature scale (see Deschamps & Sotin, 2001, for details) and δ is given by the TBL Rayleigh number, the presence of an insulating crust within the stagnant lid does not affect the value of heat flux in such an approach. However, this study demonstrates that the presence of the clathrate crust increases the convective temperature and thus ΔT_{δ} cannot be evaluated simply from the viscous temperature scale. If one wants to use scaling laws, an additional constraint has to be added to address the heat transfer through the insulating crust. Previous studies on ocean worlds (Tobie et al., 2006 for Titan, Kamata et al., 2019 for Pluto) used a different approach in adapting the standard scaling laws to the presence of an insulator in the stagnant lid. While Tobie et al. (2006) found that the amount of extracted internal heat is significantly reduced, Kamata et al. (2019) reported an opposite result noting that the insulating surface layer enhances thermal convection in the ice shell. This disagreement shows the necessity of performing dedicated numerical simulations. Our study which is the first to present numerical simulations with the clathrate crust in the stagnant lid agrees with the findings of Tobie et al. (2006). Scaling laws have also been adapted to assess the effect of a low-conductivity regolith on the thermal evolution of Mars (Schumacher & Breuer, 2006; Thiriet et al., 2018). These authors found that the presence of an insulating surface layer reduces the stagnant lid thickness and limits the cooling rate of the Martian interior. The present study is in qualitative agreement with their findings.

Our results show that the clathrate crust significantly limits heat extraction and thus its presence enables a long-term stability of the deep ocean. Alternatively, an ammonia-rich ocean has been proposed because ammonia lowers the freezing point and, consequently, the convective ice temperature. This leads to higher viscosity which also limits the amount of transferred heat (Grasset & Sotin, 1996). The presence of ammonia would, however, significantly reduce the ocean density which seems to be ruled out by the large value of the k_2 Love number (Durante et al., 2019; Mitri et al., 2014). Recently, Leitner and Lunine (2019) also proposed that the current ammonia concentration in Titan's ocean should not exceed 5%. For a 100 km thick ice shell, this would only decrease the ocean temperature by less than 5 K—probably not enough to sufficiently reduce the amount of extracted heat. On other icy moons, such as Enceladus and Europa, the presence of a deep ocean is best explained by tidal dissipation. In the case of Titan, the amount of tidal dissipation is much lower owing to its large distance from Saturn (Tobie, Grasset, et al., 2005; Tobie, Mocquet, et al., 2005). In addition, significant tidal dissipation would circularize the orbit which is difficult to reconcile with Titan's large eccentricity. Our results show that the insulating clathrate crust provides a plausible alternative to explain the maintenance of Titan's deep ocean.

A thinner stagnant lid may facilitate the transfer of organics from the surface to the interior. On the Earth, water alters rock down to 8 km below the sea floor (Andreani et al., 2007). The penetration depth of liquids is controlled by tectonics and pressure. With gravity on Titan being more than 7 times lower than on the Earth and the water ice or clathrate density being at least 2 times smaller than that of the terrestrial crust, one can expect fluids to percolate as deep as a few tens of kilometers although Titan's tectonics seems much less active. If the organic compounds present on Titan's surface (e.g., acetylene) can dissolve in the hydrocarbons (Singh et al., 2017), they could be entrained by the percolating liquids. Since the clathrate crust significantly reduces the stagnant lid thickness, the organics entrained by liquids may reach the convective domain and could be carried downward to the ocean. Such a transfer of organic compounds from the surface to the ocean may have profound astrobiological implications. Other processes that could contribute to the transfer of organics from the surface to the convective layer include fractures that may propagate to the brittle-ductile transition or impacts that could break through the thinner stagnant lid down to the convective domain.

Impacts on a clathrate subsurface can also deliver methane into the atmosphere. Events that created craters such as Sinlap would release $\sim 1\%$ of the present amount of atmospheric methane if the shell has a 5 km thick

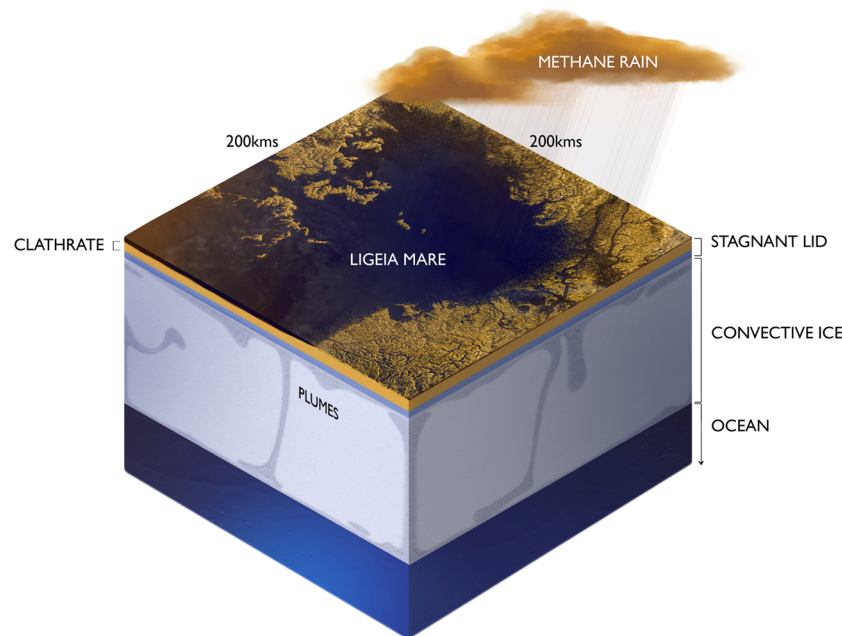


Figure 4. The clathrate crust affects Titan's climate as well as its interior dynamics. It serves as a reservoir of methane that drives Titan's weather, and it prevents the deep ocean from freezing. Its presence also dramatically reduces the thickness of the stagnant lid. The hot and cold plumes reach depths of only a few kilometers below the clathrate crust thus enhancing the potential for the exchange between the surface and the deep ocean.

clathrate crust (Choukroun & Sotin, 2012). There are a dozen “almost certain” impact craters including Menrva which is 5 times as large as Sinlap (Werynski et al., 2019). This mechanism can contribute to the replenishment of methane in the atmosphere in addition to potential cryovolcanism (Lopes et al., 2013) and the substitution of methane by ethane in polar regions (Choukroun & Sotin, 2012).

Low thermal conductivity layers have important implications for geological evolution of Titan (Tobie et al., 2006) as well as other planetary bodies. For example, an insulating CO₂ ice layer as thick as 10 km may be present on Triton (O'Hara & Dombard, 2019). A clathrate layer has also recently been proposed to explain the putative presence of Pluto's ocean (Kamata et al., 2019) as well as the high temperatures close to the surface of Ceres (Castillo-Rogez et al., 2019). Finally, clathrates have been invoked to trap oxidized molecules in Europa's ice shell (Hand et al., 2006). Formation of clathrates on top or within the outer ice shell thus does not seem to be unique to Titan. Owing to their different physical properties when compared with water ice, the presence of clathrates has profound implications on the structure and dynamics of ocean world's ice shells.

The presence of a clathrate crust capping Titan's ice shell has long been inferred (Lewis, 1971; Loveday et al., 2001; Lunine & Stevenson, 1987; Mousis et al., 2014; Tobie et al., 2006). In this study, we show that due to its low thermal conductivity, the clathrate crust can explain the persistence of Titan's deep ocean and provide a potential pathway for the exchange of material between the surface and the ocean. Future studies should address the effect of clathrate crust on the relaxation rate of impact craters, the effect of lateral thickness variations of the crust on the global energy balance, and the stability of ethane clathrates that may form at the poles (Choukroun & Sotin, 2012). The methane cycle on Titan (Hayes et al., 2018) is often compared to the water cycle on the Earth (Lorenz & Sotin, 2010) where the interaction between the ocean and the lithosphere results in the formation of weak hydrous phases that lubricate faults and lower lithospheric yield stress (Foley & Driscoll, 2016). The Earth's water cycle thus plays a major role in the dynamics of the Earth's mantle. This study demonstrates that on Titan, methane plays a similar role and can have a crucial effect on the dynamics of its ice shell (Figure 4). Future missions to Titan, such as the recently selected New Frontiers mission Dragonfly (Turtle et al., 2017), will investigate the interior structure of this world that undergoes processes so similar to those on the Earth. In particular, measuring the thickness of the clathrate crust, the ice shell, and the ocean would provide strong constraints on models that investigate Titan's habitability.

Data Availability Statement

All data used for producing figures in this paper can be found online (<https://doi.org/10.5281/zenodo.3638226>).

Acknowledgments

The authors are grateful to two reviewers for their comments that helped to improve the clarity of the manuscript. We would also like to thank Ondřej Čadek and Gabriel Tobie for fruitful discussions on tidal dissipation in Titan's shell. K. K. was supported by the Czech Science Foundation through Project 19-10809S and by Charles University Research Program UNCE/SCI/023. This research was also supported by The Ministry of Education, Youth and Sports from the Large Infrastructures for Research, Experimental Development and Innovations project "IT4Innovations National Supercomputing Center—LM2015070." Part of this work was carried out at the Jet Propulsion Laboratory, California Institute of Technology, under a contract with the National Aeronautics and Space Administration. C. S. acknowledges support by the NASA Astrobiology Institute through project "Habitability of Hydrocarbon Worlds: Titan and Beyond" (17-NAI8_2-0017).

References

- Alnæs, M., Blechta, J., Hake, J., Johansson, A., Kehlet, B., Logg, A., et al. (2015). The FEniCS Project Version 1.5. *Archive of Numerical Software*, 3. <https://doi.org/10.11588/ans.2015.100.20553>
- Andreani, M., Mével, C., Boullier, A.-M., & Escartin, J. (2007). Dynamic control on serpentine crystallization in veins: Constraints on hydration processes in oceanic peridotites. *Geochemistry, Geophysics, Geosystems*, 8, Q02012. <https://doi.org/10.1029/2006GC001373>
- Baland, R.-M., Tobie, G., Lefevre, A., & Hoolst, T. V. (2014). Titan's internal structure inferred from its gravity field, shape, and rotation state. *Icarus*, 237, 29–41. <https://doi.org/10.1016/j.icarus.2014.04.007>
- Béghin, C., Randriamboarison, O., Hamelin, M., Karkoschka, E., Sotin, C., Whitten, R. C., et al. (2012). Analytic theory of Titan's Schumann resonance: Constraints on ionospheric conductivity and buried water ocean. *Icarus*, 218(2), 1028–1042. <https://doi.org/10.1016/j.icarus.2012.02.005>
- Braukmüller, N., Wombacher, F., Hezel, D. C., Escoube, R., & Mu'ner, C. (2018). The chemical composition of carbonaceous chondrites: Implications for volatile element depletion, complementarity and alteration. *Geochimica et Cosmochimica Acta*, 239, 17–48. <https://doi.org/10.1016/j.gca.2018.07.023>
- Castillo-Rogez, J. C., Hesse, M. A., Formisano, M., Sizemore, H., Bland, M., Ermakov, A. I., & Fu, R. R. (2019). Conditions for the long-term preservation of a deep brine reservoir in Ceres. *Geophysical Research Letters*, 46, 1963–1972. <https://doi.org/10.1029/2018GL081473>
- Chen, E., Nimmo, F., & Glatzmaier, G. (2014). Tidal heating in icy satellite oceans. *Icarus*, 229, 11–30. <https://doi.org/10.1016/j.icarus.2013.10.024>
- Choblet, G., Tobie, G., Sotin, C., Běhounková, M., Čadek, O., Postberg, F., & Souček, O. (2017). Powering prolonged hydrothermal activity inside Enceladus. *Nature Astronomy*, 12, 841–847. <https://doi.org/10.1038/s41550-017-0289-8>
- Choukroun, M., & Sotin, C. (2012). Is Titan's shape caused by its meteorology and carbon cycle? *Geophysical Research Letters*, 39, L04201. <https://doi.org/10.1029/2011GL050747>
- Deschamps, F., & Sotin, C. (2001). Thermal convection in the outer shell of large icy satellites. *Journal of Geophysical Research*, 106(E3), 5107–5121. <https://doi.org/10.1029/2000JE001253>
- Durante, D., Hemingway, D., Racioppa, P., Iess, L., & Stevenson, D. (2019). Titan's gravity field and interior structure after Cassini. *Icarus*, 326, 123–132. <https://doi.org/10.1016/j.icarus.2019.03.003>
- Foley, B. J., & Driscoll, P. E. (2016). Whole planet coupling between climate, mantle, and core: Implications for rocky planet evolution. *Geochemistry, Geophysics, Geosystems*, 17, 1885–1914. <https://doi.org/10.1002/2015GC006210>
- Goldsby, D. L., & Kohlstedt, D. L. (2001). Superplastic deformation of ice: Experimental observations. *Journal of Geophysical Research*, 106(B6), 11,017–11,030. <https://doi.org/10.1029/2000JB900336>
- Grasset, O., & Sotin, C. (1996). The cooling rate of a liquid shell in Titan's interior. *Icarus*, 123(1), 101–112. <https://doi.org/10.1006/icar.1996.0144>
- Hand, K. P., Chyba, C. F., Carlson, R. W., & Cooper, J. F. (2006). Clathrate hydrates of oxidants in the ice shell of Europa. *Astrobiology*, 6(3), 463–482. <https://doi.org/10.1089/ast.2006.6.463>
- Hayes, A., Aharonson, O., Callahan, P., Elachi, C., Gim, Y., Kirk, R., et al. (2008). Hydrocarbon lakes on Titan: Distribution and interaction with a porous regolith. *Geophysical Research Letters*, 35, L09204. <https://doi.org/10.1029/2008GL033409>
- Hayes, A. G., Lorenz, R. D., & Lunine, J. I. (2018). A post-Cassini view of Titan's methane-based hydrologic cycle. *Nature Geoscience*, 11, 306–313. <https://doi.org/10.1038/s41561-018-0103-y>
- Hobbs, P. V. (1974). *Ice physics* (pp. 837). Oxford: Clarendon Press.
- Iess, L., Jacobson, R. A., Ducci, M., Stevenson, D. J., Lunine, J. I., Armstrong, J. W., et al. (2012). The tides of Titan. *Science*, 337(6093), 457–459. <https://doi.org/10.1126/science.1219631>
- Kamata, S., Nimmo, F., Sekine, Y., Kuramoto, K., Noguchi, N., Kimura, J., & Tani, A. (2019). Pluto's ocean is capped and insulated by gas hydrates. *Nature Geoscience*, 12, 407–410. <https://doi.org/10.1038/s41561-019-0369-8>
- Leitner, M. A., & Lunine, J. I. (2019). Modeling early Titan's ocean composition. *Icarus*, 333, 61–70. <https://doi.org/10.1016/j.icarus.2019.05.008>
- Lewis, J. S. (1971). Satellites of the outer planets: Their physical and chemical nature. *Icarus*, 15(2), 174–185. [https://doi.org/10.1016/0019-1035\(71\)90072-8](https://doi.org/10.1016/0019-1035(71)90072-8)
- Logg, A., Mardal, K.-A., & Wells, G. (2012). *Automated solution of differential equations by the finite element method, lecture notes in computational science and engineering* (Vol. 84). Springer-Verlag Berlin Heidelberg. <https://doi.org/10.1007/978-3-642-23099-8>
- Lopes, R. M. C., Kirk, R. L., Mitchell, K. L., LeGall, A., Barnes, J. W., Hayes, A., et al. (2013). Cryovolcanism on Titan: New results from Cassini RADAR and VIMS. *Journal of Geophysical Research: Planets*, 118, 416–435. <https://doi.org/10.1002/jgre.20062>
- Lorenz, R., & Sotin, C. (2010). The moon that would be a planet. *Scientific American*, 302(3), 36–43. <https://doi.org/10.1038/scientificamerican0310-36>
- Loveday, J. S., Nemes, R. J., Guthrie, M., Belmonte, S. A., Allan, D. R., Klug, D. D., et al. (2001). Stable methane hydrate above 2 GPa and the source of Titan's atmospheric methane. *Nature*, 410, 661–663. <https://doi.org/10.1038/35070513>
- Lunine, J. I., & Stevenson, D. J. (1987). Clathrate and ammonia hydrates at high pressure: Application to the origin of methane on Titan. *Icarus*, 70(1), 61–77. [https://doi.org/10.1016/0019-1035\(87\)90075-3](https://doi.org/10.1016/0019-1035(87)90075-3)
- Mastrogiuseppe, M., Hayes, A., Poggiali, V., Seu, R., Lunine, J. I., & Hofgartner, J. D. (2016). Radar sounding using the Cassini altimeter: Waveform modeling and Monte Carlo approach for data inversion of observations of Titan's seas. *IEEE Transactions on Geoscience and Remote Sensing*, 54(10), 5646–5656. <https://doi.org/10.1109/TGRS.2016.2563426>
- Miller, K. E., Glein, C. R., & Waite, J. H. Jr. (2019). Contributions from accreted organics to Titan's atmosphere: New insights from cometary and chondritic data. *The Astrophysical Journal*, 871, 59. <https://doi.org/10.3847/1538-4357/aaf561>
- Mitchell, J. L., & Lora, J. M. (2016). The climate of Titan. *Annual Review of Earth and Planetary Sciences*, 44(1), 353–380. <https://doi.org/10.1146/annurev-earth-060115-012428>
- Mitri, G., Meriggiola, R., Hayes, A., Lefevre, A., Tobie, G., Genova, A., et al. (2014). Shape, topography, gravity anomalies and tidal deformation of Titan. *Icarus*, 236, 169–177. <https://doi.org/10.1016/j.icarus.2014.03.018>

- Mousis, O., Choukroun, M., Lunine, J. I., & Sotin, C. (2014). Equilibrium composition between liquid and clathrate reservoirs on Titan. *Icarus*, 239, 39–45. <https://doi.org/10.1016/j.icarus.2014.05.032>
- Néri, A., Guyot, F., Reynard, B., & Sotin, C. (2020). A carbonaceous chondrite and cometary origin for icy moons of Jupiter and Saturn. *Earth and Planetary Science Letters*, 530(115), 920. <https://doi.org/10.1016/j.epsl.2019.115920>
- O'Hara, S., & Dombard, A. (2019). Simulating formation of Triton's cantaloupe terrain by compositional diapirs, in lunar and planetary science conference. *Lunar and Planetary Science Conference*, 50, 2089.
- Schneider, T., Graves, S. D. B., Schaller, E. L., & Brown, M. E. (2012). Polar methane accumulation and rainstorms on Titan from simulations of the methane cycle. *Nature*, 481, 58–61. <https://doi.org/10.1038/nature10666>
- Schumacher, S., & Breuer, D. (2006). Influence of a variable thermal conductivity on the thermochemical evolution of Mars. *Journal of Geophysical Research*, 111, E02006. <https://doi.org/10.1029/2005JE002429>
- Singh, S., Combe, J.-P., Cordier, D., Wagner, A., Chevrier, V., & McMahon, Z. (2017). Experimental determination of acetylene and ethylene solubility in liquid methane and ethane: Implications to Titan's surface. *Geochimica et Cosmochimica Acta*, 208, 86–101. <https://doi.org/10.1016/j.gca.2017.03.007>
- Sloan, E., Koh, C., & Koh, C. (2007). *Clathrate hydrates of natural gases*. Chemical Industries, CRC Press.
- Stofan, E. R., Elachi, C., Lunine, J. I., Lorenz, R. D., Stiles, B., Mitchell, K. L., et al. (2007). The lakes of Titan. *Nature*, 445, 61–64. <https://doi.org/10.1038/nature05438>
- Thiriet, M., Michaut, C., Breuer, D., & Plesa, A.-C. (2018). Hemispheric dichotomy in lithosphere thickness on Mars caused by differences in crustal structure and composition. *Journal of Geophysical Research: Planets*, 123, 823–848. <https://doi.org/10.1002/2017JE005431>
- Tobie, G., Choblet, G., & Sotin, C. (2003). Tidally heated convection: Constraints on Europa's ice shell thickness. *Journal of Geophysical Research*, 108(E11), 5124. <https://doi.org/10.1029/2003JE002099>
- Tobie, G., Grasset, O., Lunine, J. I., Mocquet, A., & Sotin, C. (2005). Titan's internal structure inferred from a coupled thermal-orbital model. *Icarus*, 175(2), 496–502. <https://doi.org/10.1016/j.icarus.2004.12.007>
- Tobie, G., Lunine, J. I., & Sotin, C. (2006). Episodic outgassing as the origin of atmospheric methane on Titan. *Nature*, 440, 61–64. <https://doi.org/10.1038/nature04497>
- Tobie, G., Mocquet, A., & Sotin, C. (2005). Tidal dissipation within large icy satellites: Applications to Europa and Titan. *Icarus*, 177(2), 534–549. <https://doi.org/10.1016/j.icarus.2005.04.006>
- Turtle, E. P., Barnes, J. W., Trainer, M. G., Lorenz, R. D., MacKenzie, S. M., Hibbard, K. E., et al. (2017). Dragonfly: Exploring Titan's prebiotic organic chemistry and habitability, in lunar and planetary science conference. *Lunar and Planetary Science Conference*, 48, 1958.
- Turtle, E. P., Perry, J. E., Barbara, J. M., Del Genio, A. D., Rodriguez, S., Le Mouélic, S., et al. (2018). Titan's meteorology over the Cassini mission: Evidence for extensive subsurface methane reservoirs. *Geophysical Research Letters*, 45, 5320–5328. <https://doi.org/10.1029/2018GL078170>
- Van Schmus, W. (2013). *Natural radioactivity of the crust and mantle* (pp. 283–291). American Geophysical Union (AGU). <https://doi.org/10.1029/RF001p0283>
- Vu, T. H., Choukroun, M., Sotin, C., Muñoz-Iglesias, V., & Maynard-Casely, H. E. (2020). Rapid formation of clathrate hydrate from liquid ethane and water ice on Titan. *Geophysical Research Letters*, 47, e2019GL086265. <https://doi.org/10.1029/2019GL086265>
- Vuitton, V., Yelle, R., & McEwan, M. (2007). Ion chemistry and N-containing molecules in Titan's upper atmosphere. *Icarus*, 191(2), 722–742. <https://doi.org/10.1016/j.icarus.2007.06.023>
- Werynski, A., Neish, C., Gall, A. L., & Janssen, M. (2019). Compositional variations of Titan's impact craters indicates active surface erosion. *Icarus*, 321, 508–521. <https://doi.org/10.1016/j.icarus.2018.12.007>
- Wilson, E. H., & Atreya, S. K. (2000). Sensitivity studies of methane photolysis and its impact on hydrocarbon chemistry in the atmosphere of Titan. *Journal of Geophysical Research*, 105(E8), 20,263–20,273. <https://doi.org/10.1029/1999JE001221>
- Yung, Y. L., & Allen, M. (1984). P. J. P Photochemistry of the atmosphere of Titan: Comparison between model and observations. *Astrophysical Journal Supplement Series*, 55, 465–506. <https://doi.org/10.1086/190963>